Zhaofu (Jeff) Chen

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University of California, Irvine
B.S. Chemical & Biomolecular Engineering; Minor in Materials Science and Engineering

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LinkedIn Profile
Expected Graduation: Jun 2027

Education

University of California, Irvine *Expected Jun 2027*

- *B.S. Chemical & Biomolecular Engineering, Henry Samueli School of Engineering; Minor in Materials Science and Engineering* Irvine, CA

Relevant coursework: Organic Chemistry; Chemical Processing & Thermodynamics; Numerical Analysis and Computer Algorithms in MATLAB; Principles of Materials Science and Engineering; Fluid Dynamics; Chemical Reaction Engineering (CRE); Heat Transfer; Mass Transfer; Separation Process; Physical Science

Publications

- **3D nanoscale design of oxide morphology, anisotropy and chemical heterogeneity by glancing angle physical vapor deposition.** Manuscript submitted to *Nanoscale*; 4th author.
- **Enhanced Third Harmonic Generation in PEDOT:PSS Polymer Films Enabled by Femtosecond Laser Irradiation.** Manuscript ready; target: *Nature* ; 3rd author.
- **Tunable Conducting Polymers for Nonlinear and Epsilon-Near-Zero Optics: A Review.** Manuscript in progress; currently revising second draft.
- **Amorphous high-entropy oxides for CO_2 electroreduction.** Manuscript in preparation; target: *Advanced Materials*; first author.
- **Solid-state synthesis & characterization of indium-oxide-based medium-entropy oxides.** Manuscript submitted to *Journal of the American Ceramic Society*; co-first author (equal contribution). (ChemRxiv preprint, 2025)
- **Advances and outlook in modified graphene oxide (GO)/epoxy composites for mechanical applications.** Guo, L., Chen, Z., et al., *Applied Nanoscience* 13, 3273–3287 (2023).

Research Experience & Funded Projects

Solid-state Synthesis & Characterization of Indium-Oxide-based Medium-Entropy Oxides *2024 – Present*

- *Composition–processing–phase stability relationships in indium-oxide-based medium-entropy oxides.*

  - **XRD** phase identification/indexing + quantitative phase analysis; built processing–phase maps to guide sintering windows.
  - Built **Python**/**MATLAB** pipelines for batch comparison, statistical analysis, and publication-quality plots/tables.
  - Publication: ChemRxiv preprint (2025); **co-first author (equal contribution)**; manuscript **submitted** to *Journal of the American Ceramic Society.*

Amorphous High-Entropy Oxides for CO_2 Electroreduction *2024 – Present*

- *Correlated amorphous structure/chemistry with electrochemical performance; emphasis on rigorous electrical/EIS workflows.*

  - **EIS**: equivalent-circuit fitting + parameter tracking (Nyquist/Bode) across compositions; summarized trends with publication-quality figures.
  - 4-probe setup: developed measurement workflow and performed extensive troubleshooting to eliminate contact artifacts (probe contact instability/contact resistance).
  - Temperature-dependent protocol: implemented temperature-ramping measurements; iteratively debugged contacts/hardware for stable ramp data.

- **XRD** phase comparison/indexing + plotting; **XPS** analysis to confirm oxygen-vacancy signatures; **SEM–EDS** experiments and figure preparation; **PLD** + sample preparation.
- Proposal/writing: primary proposal lead; **co-first author (equal contribution)**; manuscript in preparation (target: *Advanced Materials*).

Dense YSZ Thin Films on Si with Preferred z-Orientation via PLD *2024 – Present*

Process–texture–property linking for oriented YSZ thin films supporting GLAD zig-zag architectures.

- **XRD** peak/texture processing (background subtraction, peak fitting, texture metrics) and automated plotting for batch comparisons.
- Publication: *3D nanoscale design of oxide morphology, anisotropy and chemical heterogeneity by glancing angle physical vapor deposition*; manuscript submitted to *Nanoscale*; 4th author.

Tuning Oxygen-Vacancy Concentrations in $InSnZnO_x$-based CCO Thin Films *2025 – Present*

*Complex concentrated oxide thin films engineered for **memristor** devices via oxygen-vacancy control.*

- Proposal: primary proposal lead (objectives, methods, milestones) and experimental roadmap for device/reliability studies.
- Data/analysis: automated 4-probe/2-probe **I–V** processing; cycle-to-cycle statistics; threshold voltage/resistance-state extraction with summary dashboards.
- **TEM/STEM-HAADF** training plan (funded): staged training at UCI IMRI on FIB sample prep (cross-section/plan-view), **STEM-HAADF** imaging, and **EDS/EELS** mapping for device-level structure–chemistry correlation.
- Quantification: MLLS fitting of ELNES (O K / Cu L) to assess oxygen-vacancy signatures; image analysis vs. p_{O_2}.
- Status: experiments ongoing (**PLD** growth + device testing); iterating deposition/processing conditions.

Tunable Conducting Polymers for Nonlinear and Epsilon-Near-Zero Optics: A Review *2026*

Review paper connecting doping/microstructure control to nonlinear/ENZ device opportunities.

- Conducted literature review: searched, screened, and organized prior work on tunable conducting polymers for nonlinear/ENZ optics.
- Supported manuscript development through figure/modeling workflow and draft revisions (currently revising second draft).

PEDOT:PSS Laser Writing & Nonlinear Microscopy (Howard Lee Lab) *2026 – Present*

*Experimental measurements and analysis for **PEDOT:PSS**-based laser writing and nonlinear microscopy.*

- Goal: enhance neural-network bioelectrical signaling using **PEDOT:PSS** electrodes; quantified conductivity changes vs. formulation.
- Led 4-probe testing + calculations: processed raw measurements, performed statistical summaries, and reported conclusions on **DBSA**-driven conductivity gains and whether **GOPS** impacts conductivity.
- Publication: *Enhanced Third Harmonic Generation in **PEDOT:PSS** Polymer Films Enabled by Femtosecond Laser Irradiation*; manuscript ready; target *Nature Materials*; 4th author.

Technical Control & Quality Intern, GoldenKeys High-tech Materials Co., Ltd. *Jul 2024 – Aug 2024*

Automation and data workflows for oxidation-catalyst development and quality control.

- Automated testing on **Opentrons FLEX**; implemented data schemas and QC checks for traceability.
- Streamlined multi-instrument processing (**BET**/**GC**/**HPLC**/particle sizing/moisture) via **MATLAB**/**Python**.

Technical Skills and Interests

Languages: **Python** (scientific scripting), **MATLAB**
Data/Analysis: **EIS** (Nyquist/Bode, equivalent-circuit fitting), 4-probe/2-probe **I–V** analysis, regression/curve fitting, batch processing, publication-quality plotting
Thin-Film Deposition: **PLD** (oxide thin films, process optimization)
Tools: **ImageJ**, **Origin**, **Adobe Illustrator**, Excel
Instrumentation: Biologic SP-200 potentiostat, **Opentrons FLEX** workstation; **XRD**, **XPS**, **SEM–EDS**, **BET**, **GC**, **HPLC**, thin-layer chromatography, IR, ^{1}H NMR, ellipsometry (analysis), **CARS** (Leica SP8 Falcon), **TEM/STEM-HAADF** (training in progress)
Current/Planned Fabrication Training: **AFM**, **EBL**, **EUV** lithography, two-photon lithography (3D printing), spin coating, drop coating, acid treatment, annealing, metal/oxide etching, liquid-metal printing (2D), deposition, femtosecond laser irradiation
Coursework: Organic Chemistry; Chemical Processing & Thermodynamics; Numerical Analysis and Computer Algorithms in MATLAB
Areas of Interest: High- and medium-entropy oxides, **memristors**, **CO_2 electroreduction**, defect chemistry, nonlinear optical microscopy, **2D materials** and **graphene**-based nanostructures

Conferences & Presentations

- **Optica Advanced Photonics Congress:** Conference submission participation (co-author).
- **SPIE Nanoscience + Engineering (Plasmonics: Design, Materials, Fabrication, Characterization, and Applications XXIV):** Co-author confirmation for submission titled *Enhanced Third Harmonic Generation in PEDOT:PSS Polymer Films Enabled by Femtosecond Laser Irradiation.*
- **Materials Research Society (MRS):** Conference submission participation.
- **UCI Engineering Council Research Fair (Nov 2025):** Poster presentation.
- **UCI IRT Symposium Presentation (Jun 2025):** Research presentation.

Positions of Responsibility

- **UROP Researcher,** Synthesis of Amorphous High Entropy Oxides for CO_2 Electroreduction — *2024 – 2025*
- **UROP Project Lead,** Designing nanostructured high-entropy materials for neuromorphic computing memristors — *2025 – 2026*
- **Research Fellow,** UCI California Institute for Telecommunications and Information Technology (CALIT2) Interdisciplinary Research Teams (IRT) Program — *2024 – Present*

Achievements

- **MSA Microscopy Society of America 2026 Undergraduate Research Scholarship Program Winner:** Awarded **$3,000** funding for advanced **TEM/STEM-HAADF** training.
- **California Institute for Telecommunications and Information Technology (CALIT2) UROP IRT Research Grant (2025–2026, Project Lead):** Awarded **$3,000** to extend independent undergraduate research on complex-oxide memristors.
- **California Institute for Telecommunications and Information Technology (CALIT2) IRT Research Grant (2024–2025):** Awarded **$3,000** for high-entropy oxide thin-film project; results presented at UROP Symposium and IRT Project Symposium.
- **Journal Publication (2023):** Co-author on peer-reviewed article in *Applied Nanoscience* on modified **GO**/epoxy composites for mechanical applications.